Simulation control

SWAT+ Documentation

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Purpose

`file.cio` is the master index. SWAT+ opens this file first. Each line tells the model which file to read for one input category. Every other input file in the project is reached through `file.cio`.

Source

- Reader: `src/readcio_read.f90`
- File-name fields and defaults: `src/input_file_module.f90`

Format

- Line 1: title line. Skipped by the reader.
- Lines 2 to 31: one line per input category. Each line starts with a category label (a free-form token that the reader reads but discards), followed by a fixed number of file-name fields whose order matches the corresponding `in_*` type in `input_file_module.f90`.
- Line 32: a single line for the optional output path.
- A field set to the literal token `null` disables the corresponding file.

The category labels written by tools like QSWAT+ and SWAT+ Editor are conventional. The reader does not validate them; only field order and count matter.

Categories and fields

The table lists every category in the order the reader expects them, the number of file-name fields, and the default file names defined in `input_file_module.f90`.

#	Category label	Fields (defaults from `input_file_module.f90`)
1	simulation	`time.sim`, `print.prt`, `object.prt`, `object.cnt`, `constituents.cs`
2	basin	`codes.bsn`, `parameters.bsn`
3	climate	`weather-sta.cli`, `weather-wgn.cli`, `pet.cli`, `pcp.cli`, `tmp.cli`, `slr.cli`, `hmd.cli`, `wnd.cli`, `atmodep.cli`
4	connect	`hru.con`, `hru-lte.con`, `rout_unit.con`, `gwflow.con`, `aquifer.con`, `aquifer2d.con`, `channel.con`, `reservoir.con`, `recall.con`, `exco.con`, `delratio.con`, `outlet.con`, `chandeg.con`
5	channel	`initial.cha`, `channel.cha`, `hydrology.cha`, `sediment.cha`, `nutrients.cha`, `channel-lte.cha`, `hyd-sed-lte.cha`, `temperature.cha`
6	reservoir	`initial.res`, `reservoir.res`, `hydrology.res`, `sediment.res`, `nutrients.res`, `weir.res`, `wetland.wet`, `hydrology.wet`
7	routing_unit	`rout_unit.def`, `rout_unit.ele`, `rout_unit.rtu`, `rout_unit.dr`
8	hru	`hru-data.hru`, `hru-lte.hru`
9	exco	`exco.exc`, `exco_om.exc`, `exco_pest.exc`, `exco_path.exc`, `exco_hmet.exc`, `exco_salt.exc`
10	recall	`recall.rec`
11	dr	`delratio.del`, `dr_om.del`, `dr_pest.del`, `dr_path.del`, `dr_hmet.del`, `dr_salt.del`
12	aquifer	`initial.aqu`, `aquifer.aqu`
13	herd	`animal.hrd`, `herd.hrd`, `ranch.hrd`
14	water_rights	`water_allocation.wro`, `element.wro`, `water_rights.wro`
15	link	`chan-surf.lin`, `aqu_cha.lin`
16	hydrology	`hydrology.hyd`, `topography.hyd`, `field.fld`
17	structural	`tiledrain.str`, `septic.str`, `filterstrip.str`, `grassedww.str`, `bmpuser.str`
18	hru_parm_db	`plants.plt`, `fertilizer.frt`, `tillage.til`, `pesticide.pes`, `pathogens.pth`, `metals.mtl`, `salt.slt`, `urban.urb`, `septic.sep`, `snow.sno`
19	ops	`harv.ops`, `graze.ops`, `irr.ops`, `chem_app.ops`, `fire.ops`, `sweep.ops`
20	lum	`landuse.lum`, `management.sch`, `cntable.lum`, `cons_practice.lum`, `ovn_table.lum`
21	chg	`cal_parms.cal`, `calibration.cal`, `codes.sft`, `wb_parms.sft`, `water_balance.sft`, `ch_sed_budget.sft`, `ch_sed_parms.sft`, `plant_parms.sft`, `plant_gro.sft`
22	init	`plant.ini`, `soil_plant.ini`, `om_water.ini`, `pest_hru.ini`, `pest_water.ini`, `path_hru.ini`, `path_water.ini`, `hmet_hru.ini`, `hmet_water.ini`, `salt_hru.ini`, `salt_water.ini`
23	soils	`soils.sol`, `nutrients.sol`, `soils_lte.sol`

#	Category label	Fields (defaults from `input_file_module.f90`)
24	decision_table	`lum.dtl`, `res_rel.dtl`, `scen_lu.dtl`, `flo_con.dtl`
25	regions	`ls_unit.ele`, `ls_unit.def`, `ls_reg.ele`, `ls_reg.def`, `ls_cal.reg`, `ch_catunit.ele`, `ch_catunit.def`, `ch_reg.def`, `aqu_catunit.ele`, `aqu_catunit.def`, `aqu_reg.def`, `res_catunit.ele`, `res_catunit.def`, `res_reg.def`, `rec_catunit.ele`, `rec_catunit.def`, `rec_reg.def`
26	pcp_path	one path string
27	tmp_path	one path string
28	slr_path	one path string
29	hmd_path	one path string
30	wnd_path	one path string
31	output_path	one path string (read as the whole line after the label)

Categories 1 to 30 are read with list-directed I/O. Category 31 (`output_path`) is read with format `(A)` and the value is taken from the trailing portion of the line after the first space.

Example

Top of `refdata/Ames_sub1/file.cio`:

```
file.cio: AMES
simulation        time.sim          print.prt         null              object.cnt        null
basin             codes.bsn         parameters.bsn
climate           weather-sta.cli   weather-wgn.cli   null              pcp.cli           tmp.cli
connect           hru.con           null              null              null              null
channel           null       null       null        null         null        null        null        null
reservoir         null       null       null        null         null        null        null        null
routing_unit      null       null       null        null
hru               hru-data.hru      null
exco              null              null              null              null              null
recall            null
dr                null              null              null              null              null
aquifer           nbull             null
```

Notes on the example:

- Line 2 (`simulation`): only `time.sim`, `print.prt`, and `object.cnt` are in use. Slots 3 and 5 (`object.prt`, `constituents.cs`) are set to `null`.
- Line 4 (`climate`): only stations, weather generator, precipitation, and temperature are active. PET, solar, humidity, wind, and atmospheric deposition are disabled.
- Line 14 (`aquifer`): the dataset writes `nbull` for the initial-aquifer file. This is the file name in this project, not a default and not a typo of `null`. The model opens whatever file name appears here.

Related files

Every input file referenced in `file.cio`. Start from the Input reference index.

Purpose

`time.sim` sets the simulation start date, end date, and the sub-daily time step. All time-driven processes (climate reads, routing, output aggregation) use these values.

Source

- Reader: src/time_read.f90
- Type definition: src/time_module.f90, `type time_current`

Format

- Line 1: title line. Skipped.
- Line 2: header line. Skipped.
- Line 3: five integer values, list-directed.

#	Name	Type	Description
1	day_start	int	Julian day of year on which the simulation starts (1 to 366)
2	yrc_start	int	Calendar year on which the simulation starts
3	day_end	int	Julian day of year on which the simulation ends
4	yrc_end	int	Calendar year on which the simulation ends
5	step	int	Sub-daily time step code (see below)

Sub-daily time step values

The type definition in `time_module.f90` documents these step codes:

Value	Meaning
0	daily
1	12-hour increment
24	hourly
96	15 minutes
1440	per minute

Defaults and derived values

The reader applies these adjustments after parsing:

- `step <= 0` is forced to `1`.
- `day_start <= 0` is forced to `1`.
- `time%nbyr` is set to `yrc_end - yrc_start + 1`.
- `time%mo`, `time%day_mo`, and `time%mo_start` are computed from `day_start` via the `xmon` helper.
- `time%yrc` is initialised to `yrc_start`.

Example

`refdata/Ames_sub1/time.sim`:

```
time.sim:
day_start  yrc_start   day_end   yrc_end      step
        0       1975         0      2020         0
```

This runs from 1 Jan 1975 through 31 Dec 2020 (the zero values for `day_start` and `day_end` are interpreted as the first and last day of the year). The `step` value of `0` is forced to `1` by the reader.

Related files

- print.prt. Controls which outputs are written within this simulation window.

Purpose

`print.prt` controls every output the model writes. It sets the print window in calendar time, the daily print interval, the average-annual print intervals, three database-output toggles, four "other output" toggles, and one row per output object with four print-frequency flags (daily, monthly, yearly, average annual).

If `print.prt` is missing or set to `null` in `file.cio`, the model still runs but writes no time-stepped output.

Source

- Reader: src/basin_print_codes_read.f90
- Type definitions: src/basin_module.f90. See `type basin_print_codes` (`pco`) and `type print_interval`.

Format

The file has fixed top-of-file blocks followed by one row per output object. There are two object-row modes, selected by the `use_obj_lbls` flag in the third block. The reader skips a header line before each block.

Block 1: window

- Line 1: title (skipped).
- Line 2: header (skipped).
- Line 3: six integers.

#	Name	Description	Default if unset
1	nyskip	number of warm-up years to skip in average-annual summaries	0
2	day_start	julian day to start daily printing	forced to 1 if 0
3	yrc_start	calendar year to start daily printing	`time%yrc` from `time.sim`
4	day_end	julian day to end daily printing	forced to 366 if 0
5	yrc_end	calendar year to end daily printing	`time%yrc + time%nbyr`
6	interval	days between daily prints	forced to 1 if `<= 0`

Block 2: average-annual print intervals

- Line 4: header (skipped).
- Line 5: `aa_numint` followed by `aa_numint` year values. If `aa_numint = 0`, only the value is read.

`aa_numint` is the number of end-years to use for average-annual outputs. When it is greater than zero, average-annual statistics are produced once for each interval defined by the listed end years.

Block 3: database output toggles

- Line 6: header (skipped).
- Line 7: three single-character flags.

Name	Values	Meaning
csvout	`n`, `y`	also write `.csv` versions of outputs
use_obj_lbls	`n`, `y`	object-row mode (see Object rows)
cdfout	`n`, `y`	also write netCDF (`.nc`) versions of outputs

Block 4: other output toggles

- Line 8: header (skipped).
- Line 9: four single-character flags.

Name	Values	Meaning
crop_yld	a, y, b, n	crop yields: a average annual, y yearly, b both, n no print
mgtout	n, y	write mgt_out.txt (management operations log)
hydcon	n, y	write hydcon.out (hydrograph connections)
fdcout	n, y	flow duration curve. Read but the type comment notes "NOT ACTIVE"

Object rows

- Line 10: header (objects daily monthly yearly avann), skipped.
- Lines 11 and following: one row per output object. Each row has five fields: a name and four flags (d, m, y, a) with values n or y.

The reader interprets the rows in one of two modes:

- use_obj_lbls = n (positional): the reader expects rows in a fixed order. The name column is read but discarded. Missing or extra rows cause silent skips of later objects when the loop hits end-of-file.
- use_obj_lbls = y (labelled): rows can be in any order. The reader matches the first column against a list of recognised names. Repeated rows for the same object are flagged via print_prt_error. Unknown names cause the model to print an error and error stop.

Recognised object names (labelled mode)

Group	Names
Basin	basin_wb, basin_nb, basin_ls, basin_pw, basin_aqu, basin_res, basin_cha, basin_sd_cha, basin_psc
Region	region_wb, region_nb, region_ls, region_pw, region_aqu, region_res, region_sd_cha, region_psc, water_allo
LSU	lsunit_wb, lsunit_nb, lsunit_ls, lsunit_pw
HRU	hru_wb, hru_nb, hru_ls, hru_pw, hru_cb, hru_cb_vars
HRU-lte	hru-lte_wb, hru-lte_nb, hru-lte_ls, hru-lte_pw
Other	channel, channel_sd, aquifer, reservoir, recall, hyd, ru, pest
Salt	basin_salt, hru_salt, ru_salt, aqu_salt, channel_salt, res_salt, wetland_salt
Constituents	basin_cs, hru_cs, ru_cs, aqu_cs, channel_cs, res_cs, wetland_cs
gwflow	gwflow_wb, gwflow_flux, gwflow_heat, gwflow_solute, gwflow_obs, gwflow_pump

Each row's four flags map to the print_interval fields d, m, y, a (daily, monthly, yearly, average annual). Value y enables that aggregation. Value n disables it.

The HRU carbon rows hru_cb and hru_cb_vars drive the legacy CSU carbon output files (alongside the standardised per-family carbon outputs). SWAT+ Editor does not write these rows; the CSU team adds them by hand. The hru_cb flag letter selects detail (l = full per-layer set with diagnostics, y = light profile set). See the carbon outputs reference.

Example

refdata/Ames_sub1/print.prt:

```
print.prt:
nyskip      day_start  yrc_start  day_end   yrc_end   interval
0           0               1975         0      2020          1
aa_int_cnt
0
csvout        use_obj_lbls  cdfout
n             n             n
```

```
crop_yld      mgtout        hydcon        fdcout
b             y             n             n
objects                  daily       monthly        yearly         avann
basin_wb                   n             n             n             y
basin_nb                   n             n             n             y
basin_ls                   n             n             n             y
basin_pw                   n             n             n             y
basin_aqu                  n             n             n             y
...
```

Window: print over 1975 to 2020, daily interval 1, no warm-up. `aa_int_cnt = 0` means a single average-annual interval over the whole period. CSV and netCDF off. Crop yields in both average-annual and yearly form. Management log on. `use_obj_lbls = n` so the rows are in fixed order.

Related files

- `time.sim`. Defines the simulation period that bounds these print windows.
- Output reference. Documents each output file produced when the corresponding row is enabled.

Purpose

`object.cnt` declares the basin name, the landscape and total areas, and how many of each spatial object type (HRU, channel, aquifer, reservoir, and so on) the project contains. The model uses these counts to allocate the `ob`, `obcs`, `obcs_alloc`, and `obom` arrays before any object data is read.

If `object.cnt` is missing or set to `null` in `file.cio`, the model prints `Cannot find object.cnt input file` and stops.

Source

- Reader: `src/basin_read_objs.f90`
- Type definitions:
  - `src/basin_module.f90`, `type basin_inputs` (`bsn`)
  - `src/hydrograph_module.f90`, `type spatial_objects` (`sp_ob`)

Format

- Line 1: title line. Skipped.
- Line 2: header line. Skipped.
- Line 3: 21 values read into `bsn` and `sp_ob`, list-directed.

#	Field	Maps to	Type	Description
1	name	`bsn%name`	char(25)	basin name (free-form)
2	ls_area	`bsn%area_ls_ha`	real	landscape area, hectares
3	tot_area	`bsn%area_tot_ha`	real	total area, hectares
4	objs	`sp_ob%objs`	int	total object count (sum of all object types)
5	hru	`sp_ob%hru`	int	number of HRUs
6	hru_lte	`sp_ob%hru_lte`	int	number of HRU-lte objects
7	ru	`sp_ob%ru`	int	number of routing units
8	gwflow	`sp_ob%gwflow`	int	number of gwflow cells
9	aqu	`sp_ob%aqu`	int	number of aquifer objects
10	chan	`sp_ob%chan`	int	number of legacy channels
11	res	`sp_ob%res`	int	number of reservoirs
12	recall	`sp_ob%recall`	int	number of recall objects
13	exco	`sp_ob%exco`	int	number of export-coefficient objects
14	dr	`sp_ob%dr`	int	number of delivery-ratio objects

#	Field	Maps to	Type	Description
15	canal	`sp_ob%canal`	int	number of canals
16	pump	`sp_ob%pump`	int	number of pumps
17	outlet	`sp_ob%outlet`	int	number of outlets
18	chandeg	`sp_ob%chandeg`	int	number of SWAT-deg channels (SWAT+ routing)
19	aqu2d	`sp_ob%aqu2d`	int	2D aquifer count (currently not used per the type comment)
20	herd	`sp_ob%herd`	int	herd count (currently not used per the type comment)
21	wro	`sp_ob%wro`	int	water-rights object count (currently not used per the type comment)

gwflow side effect

After reading, if `bsn_cc%gwflow == 1` (set via `codes.bsn`), the reader looks for `gwflow.chancells`. If that file exists and contains river cells, the reader:

- sets `sp_ob%gwflow` to the number of river cells,
- recomputes `sp_ob%objs` to `sp_ob%objs + nriv - sp_ob%aqu`,
- sets `in_con%gwflow_con` to `gwflow.con`,
- sets `sp_ob%aqu` to 0 and `in_con%aqu_con` to `null`.

If `bsn_cc%gwflow == 1` but `gwflow.chancells` is missing or empty, gwflow is silently turned off.

Example

`refdata/Ames_sub1/object.cnt`:

```
object.cnt:
name                    ls_area      tot_area       obj       hru      lhru       rtu       mfl       aqu
demo                       1.           1            12         12         0         0         0         0
```

Notes on the example:

- `name = demo`, `ls_area = 1` ha, `tot_area = 1` ha. The fields are floats; the trailing `.` makes that explicit in this dataset.
- `objs = 12` and `hru = 12`. Every spatial object in this project is an HRU.
- The header uses shortened labels (`obj`, `lhru`, `rtu`, `mfl`, `dlr`, `can`, `pmp`, `out`, `lcha`, `hrd`). These are tool conventions; the reader does not parse the header.

Related files

- `codes.bsn`. Sets `bsn_cc%gwflow`, which triggers the gwflow side effect above.
- `file.cio`. The `simulation` category points to `object.cnt`.
- Object-specific connectivity files (`hru.con`, `chandeg.con`, `aquifer.con`, `reservoir.con`, …). These must agree with the counts in `object.cnt`.